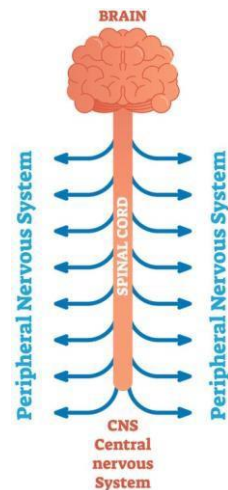


Nervous system

- **Definition:** It is the system that receives and responds to the internal and external environments in order to maintain the normal body homeostasis through controlling all functions of the body systems.
- Nervous tissue consists of two basic **types of cells:** **neurons** and **glial cells**.
- The **neuron** is the structural and functional unit of the nervous system.
- A **glial cell** is one of a variety of cells that provide a framework of tissue that supports the neurons.
- **The function of the nervous system**
 - 1) Receives the sensory inputs from all the body (**sensory functions**).
 - 2) Transmits these inputs to specific sensory areas present in the central nervous system to understand these sensations.
 - 3) Processing of these sensory inputs (**integration function**).
 - 4) Responds to these sensory inputs by motor impulse to the corresponding areas of the body (**motor functions**).
- **Classification of the nervous system**
 - ❖ **Anatomically** it is classified into:
 1. **Central nervous system (CNS)**, which lies **inside** cranial cavity or vertebral canal and consists of:
 1. Brain (inside the cranial cavity)
 2. Spinal cord (inside the vertebral canal)The brain and spinal cord are covered by protective membrane called meninges.
 2. **Peripheral nervous system (PNS)**, It lies **mostly outside** the cranial cavity or vertebral canal and consists of:
 1. Twelve (12) pairs of cranial nerves
 2. Thirty one (31) pairs of spinal nerves
 - ❖ **Functionally** the nervous system is classified into:
 1. **Somatic nervous system (SNS)**
 - It is the part of nervous system which controls all voluntary movements (حركات ارادية) and; it deals with the changes in the **external** environment.
 - It controls conscious actions whether sensory or motor (ex. Sensation of pain, touch and temperature from the skin, vision and hearing).
 - It supplies the skin, eye, ear and voluntary muscles.



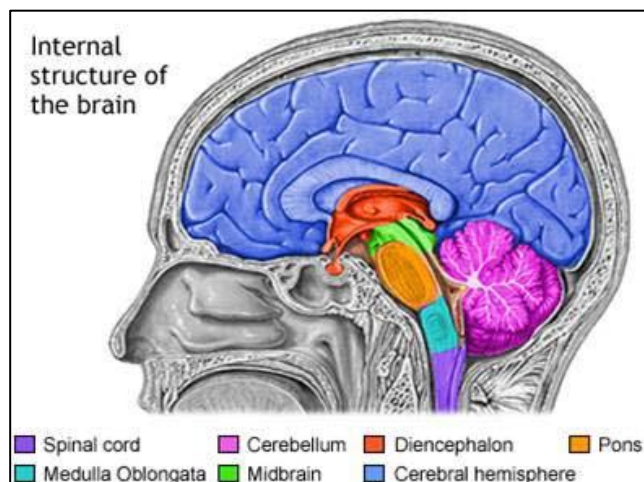
2. Autonomic nervous system (ANS)

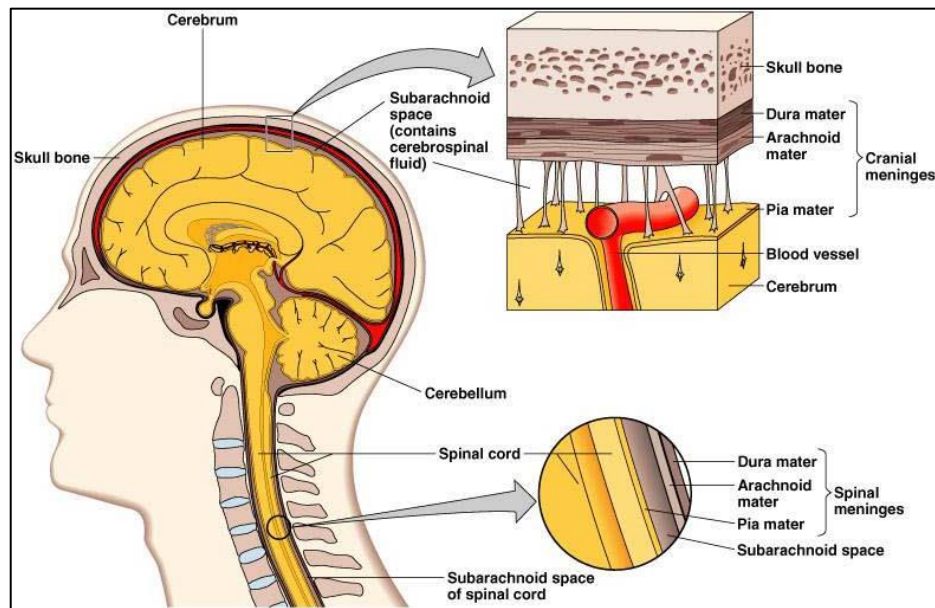
- It controls unconscious actions; regulates involuntary physiologic processes including heart rate, blood pressure, respiration, digestion.
- It deals with the changes in the **internal** environment.
- It supplies the body glands and smooth and cardiac muscles.
- It consists of two parts:
 1. **Sympathetic:** It controls the body systems under stress condition (as during muscular exercise)
 2. **Parasympathetic:** It controls the body systems under resting condition (as during sleep or relaxation الاسترخاء), secretion of glands (for digestion) and excretion (for urination and defecation).

CENTRAL NERVOUS SYSTEM (CNS)

Brain

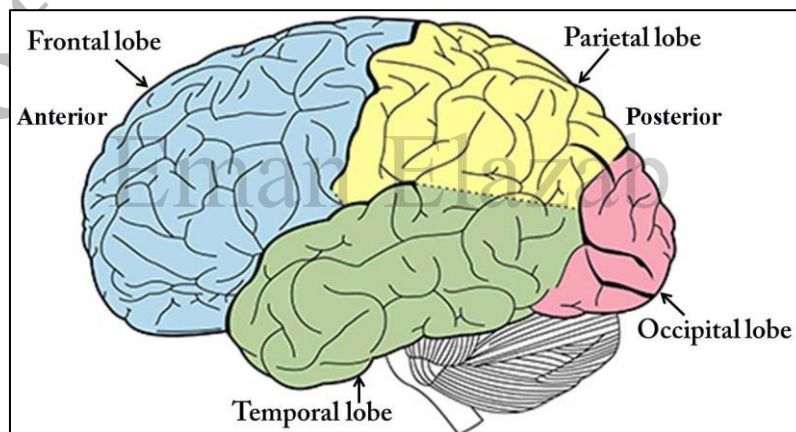
- **Site:** It lies inside the skull (cranial cavity).
- **Coverings**
 - The brain and spinal cord (CNS) are covered by the following three meninges (from outward to inward): **Dura**, **arachnoid**, and **pia** maters.
 - The pia mater is adherent to the surface of the brain and the spinal cord.
 - The space between the arachnoid mater and the pia mater is called **subarachnoid space** and is filled with cerebro-spinal fluid (CSF).
 - A potential space known as **subdural space** is present between the arachnoid and the dura mater.
- **Structure:** The brain is formed of:
 1. Two cerebral hemispheres (or cerebrum).
 2. Cerebellum
 3. Brain stem: It is subdivided into; **mid brain**, **pons** and **medulla oblongata**.





Cerebral hemispheres (Cerebrum)

- There are two **cerebral hemispheres**
- Each cerebrum is formed of four lobes: each lobe has dominant function:
 - 1. Frontal lobe** (motor functions)
 - 2. Parietal lobe** (sensory function)
 - 3. Temporal lobe** (hearing, smell, taste)
 - 4. Occipital lobe** (vision)



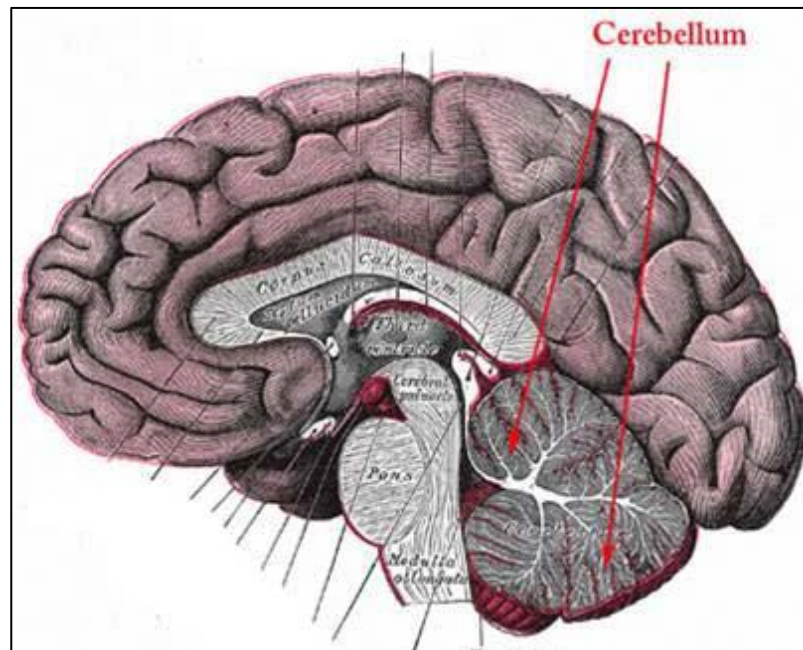
Cerebellum

- **Site**

It lies inside the cranial cavity, below the posterior part of the two cerebral hemispheres.

- **Functions**

1. Regulation of **muscle tone**.
2. Maintenance of **equilibrium**.
3. Coordination of **voluntary movements**.



Brain stem

- **Site**

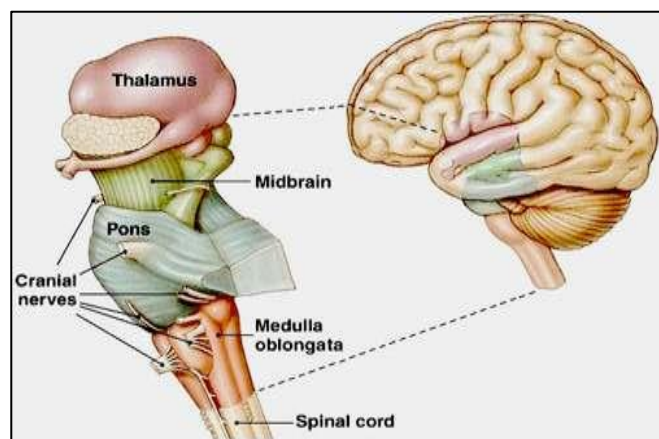
It lies in the base of the brain, and continues downwards as the spinal cord.

- **Parts:** It is formed of 3 parts; midbrain, pons, and medulla oblongata.

- **Function**

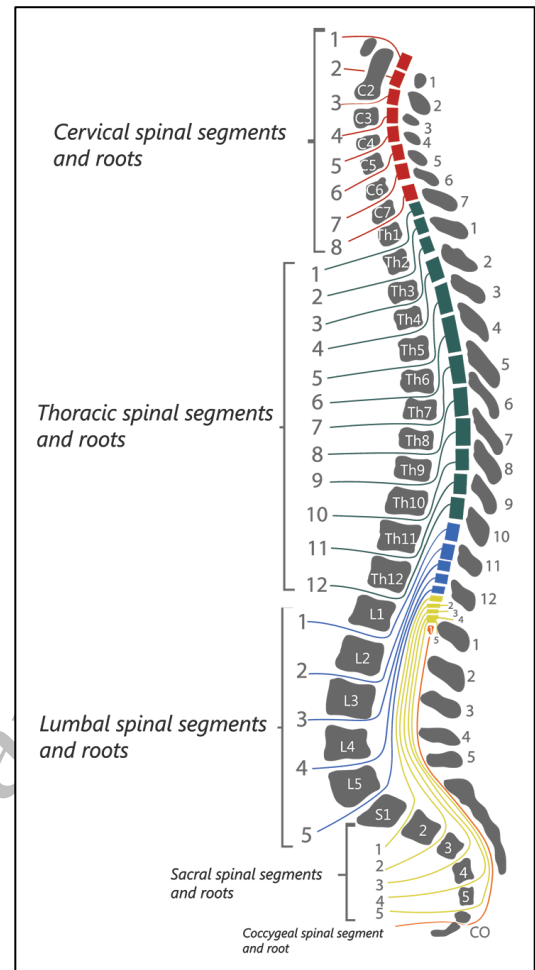
- It contains the following centers:

- 1) Respiratory center.
- 2) Cardiovascular center.
- 3) Nuclei of cranial nerves.



Spinal cord

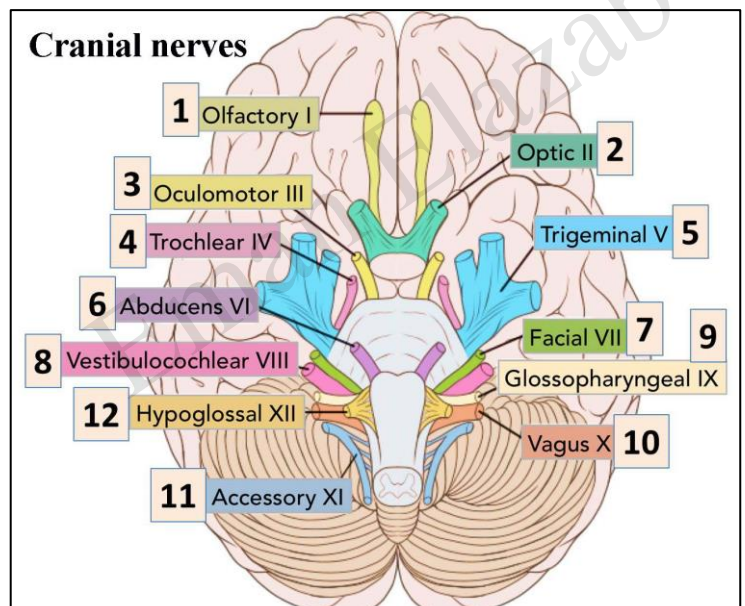
- **Site:** It lies inside the vertebral canal of the vertebral column.
- **Length:** 45 cm
- **Beginning:** In adults, it begins as the continuation of the medulla oblongata, at the **foramen magnum**.
- **Termination**
 - In adults, it terminates at the **lower border of 1st lumbar vertebra (L1)**.
 - **Coverings:** as in the brain
- **Function**
 - It gives origin for the spinal and autonomic nerves.
- **Spinal cord segments**
 - The part of spinal cord which gives attachment to a pair of spinal nerves is known as **spinal cord segment**.
 - There are 31 pairs of spinal nerves; hence, there are **31 spinal cord segments**.
 - These are: 8 cervical segments
12 thoracic segments
5 lumbar segments
5 sacral segments
1 coccygeal segment



PERIPHERAL NERVOUS SYSTEM

Cranial nerves

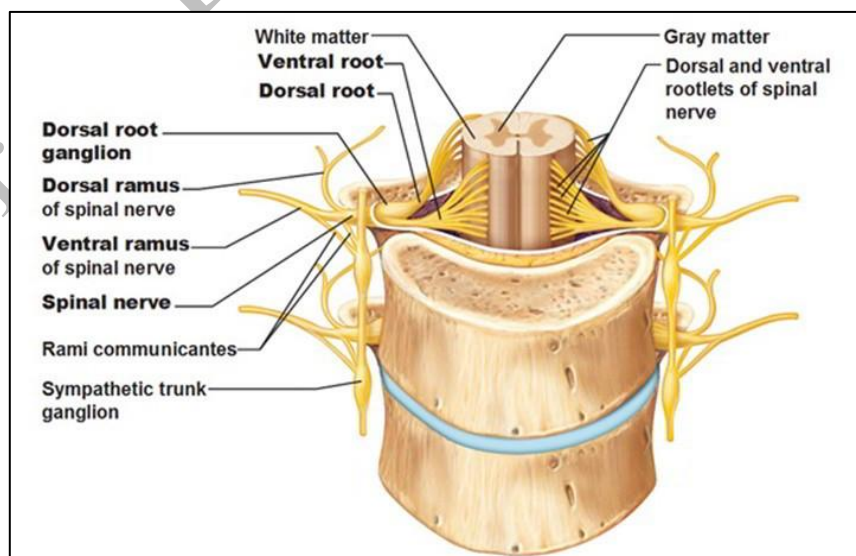
- They are 12 pairs of nerves.
- They are attached to the brain.



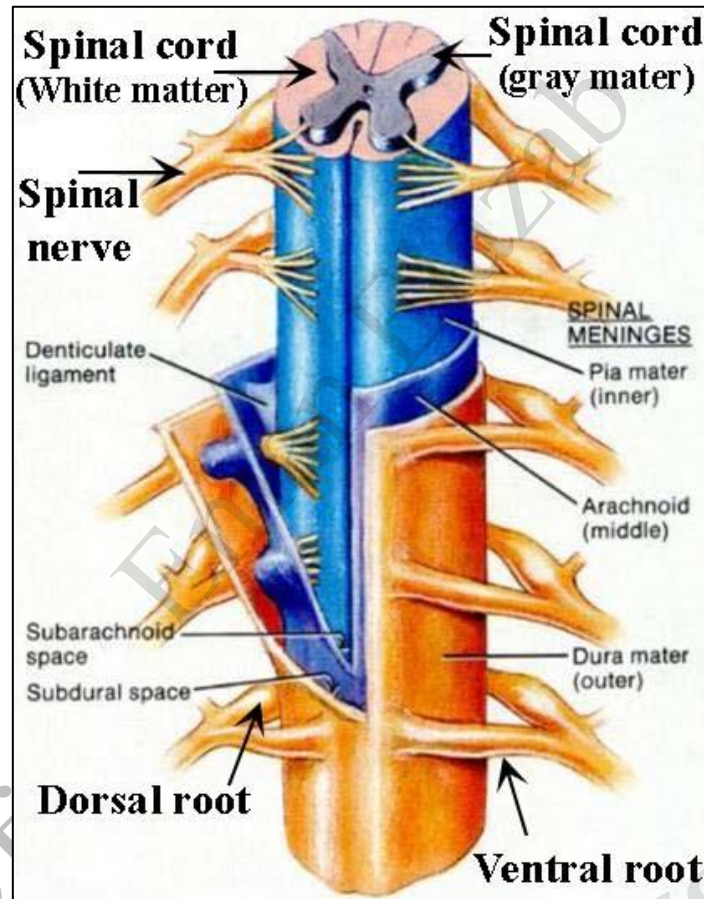
Number	Name	Functions
1	Olfactory	Smell
2	Optic	Vision
3	Oculomotor	Muscles of the eye
4	Trochlear	Muscles of the eye
5	Trigeminal	Face and teeth sensation, muscles of mastication
6	Abducent	Muscles of the eye
7	Facial	Muscles of the face
8	Vestibulocochlear	Hearing and equilibrium
9	Glossopharyngeal	Tongue and pharynx
10	Vagus	Larynx, some thoracic and abdominal organs
11	Accessory	Muscles of the neck
12	Hypoglossal	Muscles of the tongue

Spinal nerves

- There are **31 pairs** of nerves attached to the spinal cord.
- Each pair of spinal nerve is attached to one spinal cord segment.
- So their number is the same as the number of the spinal cord segments.
- Each spinal nerve is attached to each spinal cord segment by **two roots**; **ventral** and **dorsal** which unite to form a spinal nerve trunk.
- After its exit (خروجه) from the vertebral canal through intervertebral foramen it divides into **two rami**; a **ventral ramus** and a **dorsal ramus**.

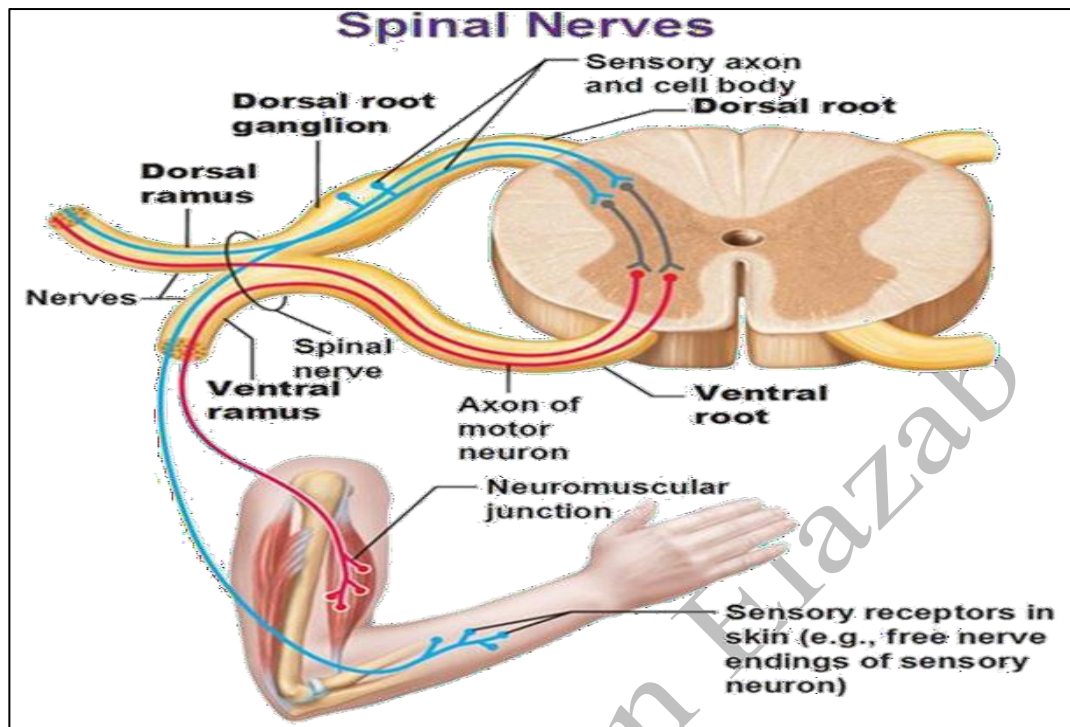


Number of Spinal cord segments	Number of Spinal nerves
8 cervical	8 cervical
12 thoracic	12 thoracic
5 lumbar	5 lumbar
5 sacral	5 sacral
1 coccygeal	1 coccygeal



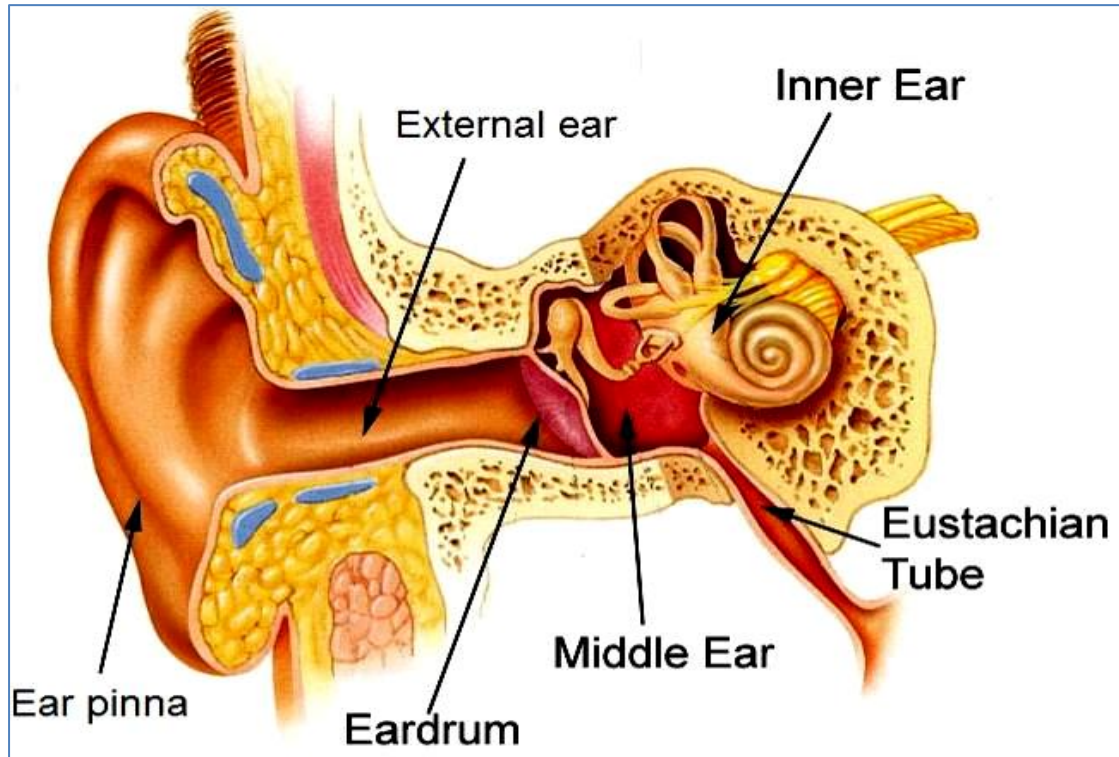
Function of peripheral nerves

- The **cranial nerves** are functionally either:
 1. Pure sensory nerve
 2. Pure motor nerve
 3. Mixed nerve (sensory and motor)
- All **spinal nerves** are mixed nerves.



Special sensory organs

THE EAR



★ It is the organ of hearing

★ It is formed of 3 parts:

I. **External ear (الأذن الخارجية):** Formed of:

A- Auricle (ear pinna).

B- External auditory meatus (قناة الأذن الخارجية).

C- Ear drum (طبلة الأذن): It separates the external ear from the middle ear.

II. **Middle ear (الأذن الوسطى):** It is a cavity inside the bone of the skull which is full of air and **contains the 3 ossicles which transmit the vibrations of sound from the external ear to the inner ear.**

- The middle ear cavity is connected to the nasopharynx by the Eustachian tube to equalize the pressure on both surfaces of the ear drum.

III. **Inner ear:** It is a cavity inside the bone of the skull which is full of fluid and **contains the sensory organs which receive the sound vibrations and transmit them to the auditory nerve.**